

Same Data, Different Destination

Thinking about a retrieval strategy for Cordwell, before we build one

Continues the Cordwell Home & Hardware scenario · Applies Modules 01 and 03 to 08

How this works. Same teams, same roles, about 45 minutes, then we reconvene. No coding, and today no schemas either. This is the most conceptual of the three exercises, and the least settled. *You are not expected to know how retrieval works.* You are expected to reason about what the data would have to look like if it did.

Pacing. Parts 1 to 3 are the core and are worth most of your time. Parts 4 and 5 can be compressed if the clock runs. Part 6 is stretch, for teams that finish early.

Everything we have built this week quietly assumed one answer to a question nobody asked out loud: how does the assistant come to know anything about Cordwell? Our answer was fine-tuning. Extract, transform, validate, profile, and load into JSONL, so the knowledge gets baked into the model's weights.

That is one answer. There is another. Today you take the same three raw sources and ask what would have to change if the assistant looked the answer up at the moment it was asked, instead of memorizing it in advance.

WHAT YOU NEED TO KNOW ABOUT RETRIEVAL TODAY, AND DELIBERATELY NO MORE

- Retrieval-augmented generation, or RAG, means this: a question arrives, you search a body of text for the most relevant passages, paste those passages into the prompt, and let the model answer from them.
- The model does not memorize your data. It reads it at answer time. Change the text, and the next answer changes.
- Retrieval operates over units of text. Something, or someone, has to decide what one unit is, what text it contains, and which facts travel alongside it. That decision is made by a data engineer, long before anything is searched.
- Everything else is Week 5. Embeddings, vector indexes, similarity scoring, top-k, re-ranking: not today. ***If your team finds itself arguing about cosine similarity, you have left the exercise.***

Today you are the data engineer, not the retrieval engineer. Every question below can be answered with what you already learned this week.

Part 1 Two answers to the same question

Before you design anything, work out what retrieval is even for. Some of these do not want retrieval at all.

For each thing Cordwell's support assistant needs to do, decide: fine-tune, retrieve, both, or neither. Yes, some are neither.

What the assistant needs to do	Fine-tune, retrieve, both, or neither?	Why
Reply in Cordwell's house voice: warm, concise, never promises a refund outright		
"Is the 20V cordless drill compatible with the battery I bought in 2019?"		
"What is Cordwell's return window for power tools?" (the policy changed last March)		
Always close by restating the ticket reference number in a fixed format		
"How do I re-thread a gas line fitting?" (long, safety-sensitive, must be exact)		
"Where is my order 48221, and has it shipped?"		

1. One row is a trap. Exactly one of these is not a knowledge problem at all, and no amount of training data or clever retrieval will ever answer it correctly. Find it, and say what the assistant should do instead.
2. State the general rule your team just discovered, in one sentence. What kind of thing does fine-tuning teach a model, and what kind of thing should be looked up instead?
3. Look back at Exercise 1. Of Cordwell's six million support tickets, which ones belong in a fine-tuning set, and which belong in a searchable corpus? Same rows, two different jobs. Can a ticket serve both?

Part 2 What, exactly, is a document?

Retrieval hands back units of text. Somebody has to define the unit. That somebody is you, and you have to do it before anyone writes a line of retrieval code.

Cordwell's three sources have very different natural shapes:

- **A support ticket.** A customer message, then an agent reply, sometimes a back and forth, plus a category and a resolution.
- **A product spec sheet.** Structured fields: dimensions, voltage, materials, compatibility list, warranty terms.
- **A live chat transcript.** Ninety minutes of interleaved conversation, three topics, two agents, one unresolved complaint.

4. For each of the three sources, what is the retrievable unit? The whole ticket? Only the agent's reply? The question and answer as a pair? One field of the spec sheet, or the whole sheet? Defend each choice, and notice that you probably did not give the same answer three times.
 5. There is a trade-off with a failure mode on each side. If the unit is too large, a fact about battery voltage sits buried inside forty lines about shipping. If it is too small, the passage "No, it is not compatible" comes back with no way to know what "it" refers to. Name both failures precisely. Which one is easier to detect after the fact?
 6. Every source has a natural seam where it can be cut without destroying meaning. Find the seam in each of the three. Then consider the ninety minute chat transcript: what breaks if you cut it every 500 characters instead of at a seam?
 7. Cordwell's spec sheets are already structured rows in a database, not prose. Does a row need to become a sentence before it can be searched by meaning? What might be gained, and what might be quietly invented, in that translation?
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Part 3 What travels with the text

A retrievable unit is not just a string. It is a record, and the fields it carries decide what your assistant is capable of.

8. The assistant answers "Yes, that drill is compatible with your battery." The customer replies, "Says who?" What fields must have travelled alongside that passage for you to answer them? List them.
 9. Cordwell's support lead asks for something reasonable: "only search tickets from the last twelve months, in the Garden category, from stores in the Southeast region." Which fields must exist on every single unit for that request to be satisfiable at all? Where do those fields live right now, and which Module 07 operation is how they get onto the ticket?
 10. Module 01 listed referential integrity as a validation concern: every foreign key points at a record that exists. Restate that requirement for a retrieval corpus. What is the concrete consequence when a retrieved passage points back at a `source_id` that no longer exists, and why is that consequence worse here than in a fine-tuning file?
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Part 4 Freshness, duplication, and what must never be retrieved

Fine-tuning bakes data in permanently. A corpus can be edited. That difference cuts in both directions.

11. Sort Cordwell's data by how fast it goes stale: product specs, return policy, prices, in-store inventory, how-to guides, resolved tickets from 2019. What does your sorting imply about how often each part of the corpus is refreshed, and what does that make retrieval good at that a fine-tuned model is bad at?
12. Module 01 asked for a duplication rate under 2%. In a fine-tuning set, duplicates cause the model to overweight whatever they say. In a retrieval corpus, near-duplicates cause a completely different failure. Work out what it is. Hint: think about what comes back when the assistant asks for the five most relevant passages and four thousand tickets say roughly the same thing.
13. In Exercise 1 your team set a policy for the customer names, addresses, and order numbers sitting in the ticket free text. Does that policy change now? Argue it both ways: an index can be scrubbed or deleted

after the fact, which a trained model cannot. But retrieved text is pasted into a prompt and read back to a human. Which risk is larger?

14. Some tickets contain an answer an agent gave that was wrong and later corrected. Some quote a return policy that has since been replaced. Both look like perfectly clean, well-formed, high-quality data. What must never enter the corpus, how would you even detect it, and is that a validation check or a human judgment?

Part 5 The contract, one more time

CALLBACK

You already have the tool for this. From Exercise 2: *“Could I answer this by looking at exactly one record, in isolation?”*

A chunk is a record. A corpus is a batch. Everything you argued out last session applies without a single change.

15. Sketch the record-level contract for one retrievable unit. Which fields must be present, and what must be true of each? Which library enforces it, and at which boundary does it run?
16. Now sketch the corpus-level invariants: the things that must be true of the collection as a whole, and that no single chunk could ever tell you. Which library enforces those?
17. Apply the diagnostic question to this one: “no two chunks in the corpus are near-duplicates of each other.” One record, or many? Which tool? And a harder follow-up, worth your remaining time: is that even expressible as a schema check, or is it something else wearing a schema's clothes? What would it cost to compute?

Part 6 Curveballs (stretch)

Pick one. None of these has a clean answer, and two of them are actively argued about in industry right now.

18. Cordwell's legal team requires that every factual claim the assistant makes can be traced to a source the customer could be shown. What does that single requirement demand of your pipeline that the fine-tuning pipeline never once asked for?
 19. Your corpus is six million tickets, and four million of them are some version of “where is my order.” Does a larger corpus make retrieval better? Make the case both ways, then decide.
 20. A teammate says: “This is overthinking it. One database row equals one chunk. Ship it.” Steelman them properly, then say precisely which of Cordwell's six needs from Part 1 would break, and how you would notice.
 21. The assistant is fine-tuned on 2024 tickets and retrieves over 2026 policy documents. Asked about returns, it confidently states the old thirty day window, ignoring the retrieved passage that says ninety. Which layer failed? Whose bug is it, and where in a pipeline would you even put a test for that?
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Report back to the group

Your reporter has about three minutes. Come ready with:

- **Your one-sentence rule** for deciding between fine-tuning, retrieving, and calling an API. All three, not two.
- **Your definition of a document** for Cordwell's support tickets, and the seam you chose to cut on.
- **One thing that must never enter the corpus**, and your honest answer on whether a schema could catch it.
- **One question you want answered in Week 5**. We will keep the list and come back to it.

Cordwell Home & Hardware is a fictional company; all data described is synthetic and for training use only.